

Agricultural shocks and the dynamics of conflict risk: Evidence from desert locust swarms - Online Appendix

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The main manuscript is available [here](#).

A Additional figures and tables

Table A1: Summary statistics

Panel A: Yearly variables

	Mean	SD	Min	50 th	75 th	Max	N
Any violent conflict event - ACLED	0.02	0.14	0.0	0.0	0.0	1.0	557018
Any violent conflict event - UCDP	0.01	0.10	0.0	0.0	0.0	1.0	557018
Any swarms in cell	0.00	0.07	0.0	0.0	0.0	1.0	557018
Any swarms within 100km outside cell	0.05	0.21	0.0	0.0	0.0	1.0	557018
Any swarms within 100-250km of cell	0.11	0.32	0.0	0.0	0.0	1.0	557018
Population (10,000s)	1.63	8.92	0.0	0.1	0.9	749.8	557018
Total annual rainfall (100 mm)	2.40	3.79	0.0	0.8	2.8	43.4	557018
Max annual temperature (deg C)	37.55	5.11	11.5	38.2	41.3	49.1	557018

Panel B: Fixed variables

	Mean	SD	Min	50 th	75 th	Max	N
Any ACLED violent conflict event in cell from 1997-2018	0.13	0.33	0.0	0.0	0.0	1.0	25435
Any UCDP violent conflict event in cell from 1997-2018	0.07	0.26	0.0	0.0	0.0	1.0	25435
Any locust swarm in cell from 1985-2021	0.20	0.40	0.0	0.0	0.0	1.0	25435
Any locust swarm in cell from 1997-2018	0.09	0.29	0.0	0.0	0.0	1.0	25435
First exposed to locust swarm between 1997-2018	0.07	0.26	0.0	0.0	0.0	1.0	25435
First exposed to locust swarm in 2003-2005 upsurge	0.05	0.22	0.0	0.0	0.0	1.0	25435
Any locust swarm within 100 km from 1985-2021	0.78	0.41	0.0	1.0	1.0	1.0	25435
Any locust swarm within 100 km from 1997-2018	0.55	0.50	0.0	1.0	1.0	1.0	25435
Any cropland or pasture in cell	0.57	0.50	0.0	1.0	1.0	1.0	25435
Share of cell allocated to crops or pasture	0.23	0.32	0.0	0.0	0.4	1.0	25435
Any pasture in cell	0.56	0.50	0.0	1.0	1.0	1.0	25435
Share of cell allocated to pasture	0.18	0.27	0.0	0.0	0.3	1.0	25435
Any cropland in cell	0.31	0.46	0.0	0.0	1.0	1.0	25435
Share of cell allocated to crops	0.05	0.13	0.0	0.0	0.0	1.0	25435

Note: Observations are grid cells approximately 28×28km by year. The study period covers 1997-2018. Data on locust swarm observations is available from the FAO Locust Watch for 1985-2021. Data on conflict events are from the ACLED and UCDP databases. Data on population is from GPW (CIESIN, 2018). Data on precipitation and temperature are from WorldClim (Fick and Hijmans, 2017). Values for land cover are for the year 2000 from Ramankutty et al. (2010).

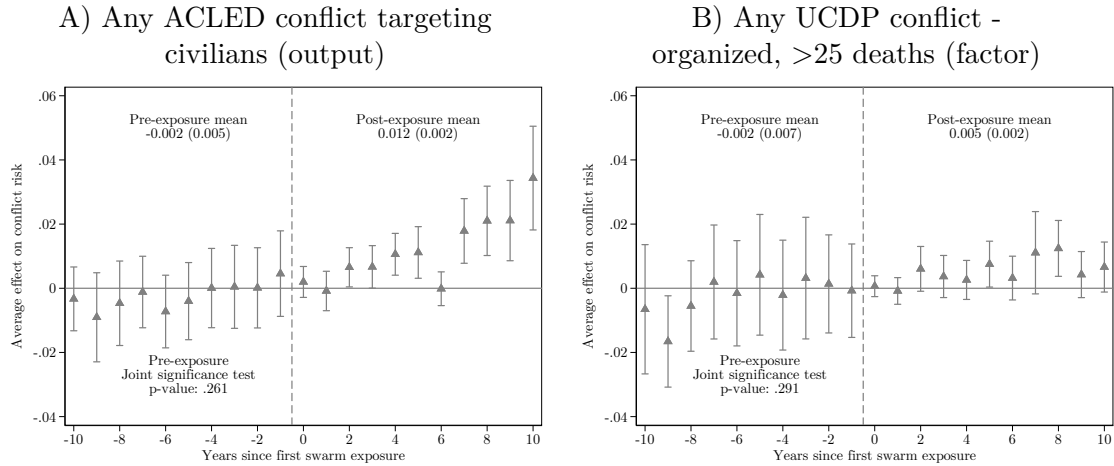
Table A2: Balance in cell characteristics by exposure to any locust swarm

	All cells		W/in 100km of any swarm		W/in 100km of any swarm	
	No weights		No weights		Inverse exposure propensity weights	
	Control	Treat	Control	Treat	Control	Treat
	Mean	Diff.	Mean	Diff.	Mean	Diff.
	(SD)	(SE)	(SD)	(SE)	(SD)	(SE)
Population in 2000 (10,000s)	1.22	1.32***	1.65	0.90***	1.67	0.43
	[6.99]	(0.37)	[8.67]	(0.33)	[8.71]	(0.50)
Mean of cell nightlights	0.05	0.01**	0.05	0.00	0.05	0.00
	[0.04]	(0.00)	[0.05]	(0.00)	[0.05]	(0.00)
Distance to national capital	707.37	-182.42***	586.82	-61.87**	586.45	17.59
(km)	[405.76]	(46.07)	[357.73]	(28.29)	[358.35]	(24.98)
Distance to nearest national	200.16	-40.97***	195.63	-36.45***	194.52	8.31
boundary (km)	[151.74]	(13.96)	[147.84]	(11.90)	[147.70]	(11.54)
Percent of cell covered by	4.69	0.83	5.83	-0.31	5.89	0.63
crop land in 2000	[13.12]	(0.78)	[14.48]	(0.69)	[14.55]	(0.80)
Percent of cell covered by	17.43	11.03***	21.59	6.87***	21.79	-0.92
pasture land in 2000	[26.69]	(2.40)	[28.05]	(2.16)	[28.12]	(1.59)
Percent of cell covered by	68.97	-9.98***	61.16	-2.16	60.76	0.98
barren area in 2009	[42.75]	(3.68)	[43.52]	(3.04)	[43.58]	(2.73)
Percent of cell covered by	0.08	0.10	0.10	0.08	0.10	0.01
urban area in 2009	[0.75]	(0.07)	[0.88]	(0.07)	[0.89]	(0.05)
Percent of cell covered by	6.80	-1.09	7.12	-1.42	7.20	-0.07
forest area in 2009	[17.00]	(1.03)	[16.15]	(0.91)	[16.22]	(0.92)
Percent of cell covered by	2.01	1.49**	2.92	0.58	2.94	0.20
water area in 2009	[10.70]	(0.61)	[12.75]	(0.61)	[12.81]	(0.53)
Mean annual max NDVI	0.23	-0.01	0.24	-0.02	0.24	0.01
(1997-2018)	[0.21]	(0.02)	[0.20]	(0.01)	[0.20]	(0.01)
Mean annual rainfall 1997-2018	2.42	0.15	2.68	-0.11	2.71	0.11
(100 mm)	[3.80]	(0.25)	[3.76]	(0.17)	[3.77]	(0.18)
Mean annual max temperature	37.63	-1.61***	36.70	-0.69*	36.67	-0.21
1997-2018 (deg C)	[5.11]	(0.45)	[5.05]	(0.37)	[5.07]	(0.38)
Mean of cell annual share of	0.09	-0.00	0.09	0.00	0.09	0.00
months with drought 1998-2014	[0.03]	(0.00)	[0.04]	(0.00)	[0.04]	(0.00)
	$F = 5.00$		$F = 2.38$		$F = 0.64$	
	$p < 0.01$		$p < 0.01$		$p = 0.824$	
Joint significance						

Note: The table shows results from separate bivariate regressions of baseline or mean cell characteristics on a dummy for being exposed to a locust swarm during the study period. The rows indicate which dependent variable is used. The first set of columns includes all cells while the second restricts the sample to cells within 100 km of any locust swarm observation from 1997-2021. The third set of columns includes all cells but weights observations by the inverse of the estimated propensity to have been exposed to a locust swarm during the study period. I include results of joint tests that there is no relationship between any of the characteristics and swarm exposure. Observations are grid cells approximately 28×28km by year. SEs clustered at the sub-national region level are in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Figure A1: Impacts of swarm exposure on conflict risk over time, by conflict type



Note: The dependent variables are dummies for any conflict event being observed in a cell in a year, with the conflict type specified in the panel title. Each panel replicates Figure 3 for a different conflict outcome. See the figure note for Figure 3 for more detail. Exact coefficients are reported in Table B1.

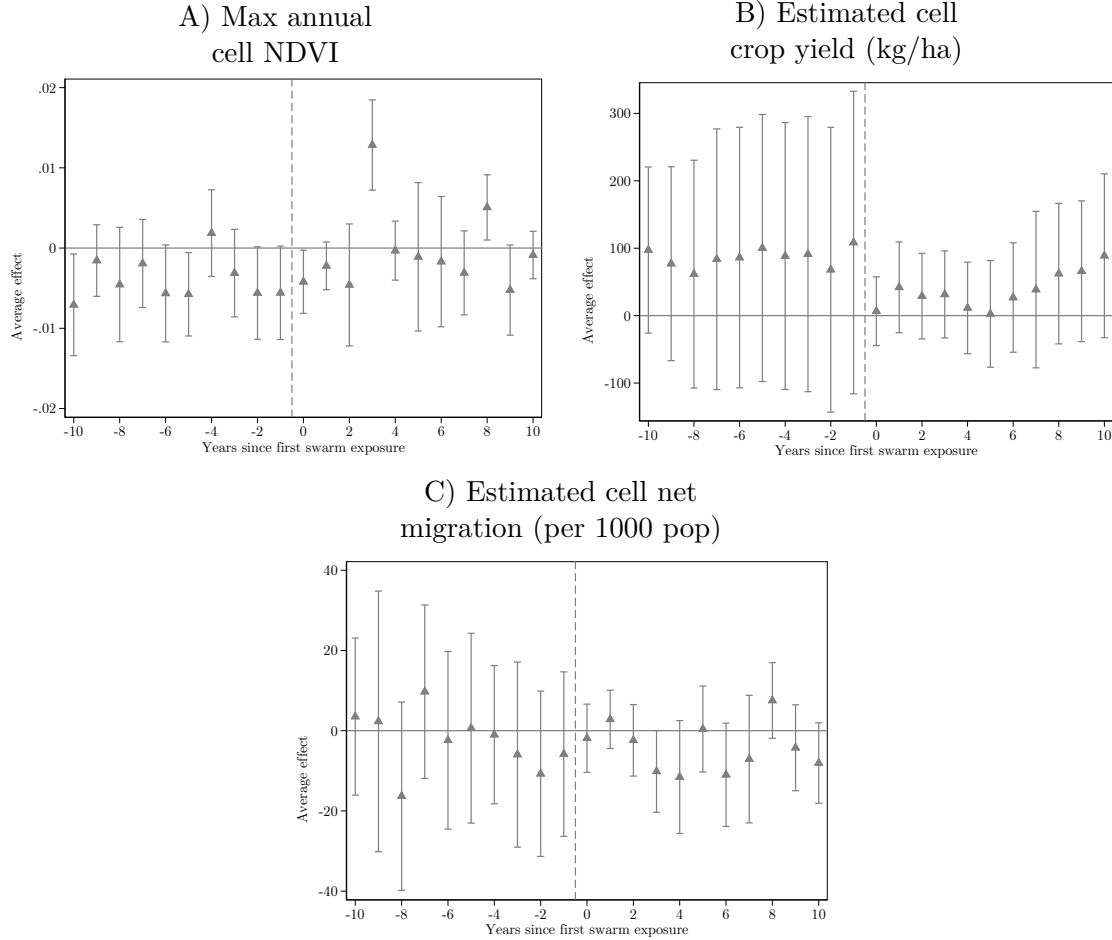
Table A3: Heterogeneity in impacts of exposure to locust swarms on violent conflict risk by exposure to other shocks

Outcome: Any violent conflict event	(1) Any active famine in country	(2) Any active drought in cell	(3) Any drought prev. year in cell	(4) Any swarm before 1990 in cell
Effect of exposure, No shock	0.016*** (0.004)	0.011*** (0.003)	0.018*** (0.004)	0.017*** (0.004)
Effect of exposure, Any shock	0.060*** (0.017)	0.015** (0.007)	0.023*** (0.008)	0.019*** (0.006)
Observations	301664	294353	295564	301664
<i>p</i> -value, test of equality of coefficients	0.010	0.511	0.527	0.626
Country-Year FE	Yes	Yes	Yes	Yes
Cell FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes

Note: The table presents results from separate regressions where I estimate heterogeneous average long-term impacts of locust swarm exposure on the probability of any violent conflict event in a cell-year by indicators of other shocks using the BJS estimator and built-in heterogeneity option in the Stata implementation of the estimator. The famine indicator in column 1 is defined at the country-year level by whether any famine was declared. The drought indicators in columns 2-3 are cell-year level variables, defined by whether there are at least 4 consecutive months where the SPEI is below -1.5. Swarm exposure prior to 1990 in column 4 considers exposure in the period from 1985-1989 where there were several major locust upsurges not considered in the definition of swarm exposure in the study sample period. At the bottom of the table I include *p*-values for the test of equality of the coefficients representing effects by the presence of civil conflict. All regressions include country-by-year and cell fixed effects and controls for annual precipitation, maximum temperature, and population, and distance to capital by year fixed effects. Observations are grid cells approximately 28×28km by year. SEs are clustered at the sub-national region level.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Figure A2: Impacts of exposure to locust swarms on economic outcomes over time



Each panel replicates [Figure 3](#) but considering a different outcome as indicated in the panel title. See the figure note for [Figure 3](#) for more detail.

NDVI is calculated from MODIS satellite imagery from 2000-2018 (Didan, [2015](#)). I calculate average NDVI in each cell-month based on 16-day NDVI observations, and then take the maximum of these values in each cell-year. The event study is restricted to crop cells, where NDVI is a potential proxy for agricultural production. Crop-specific yield data at the cell level are estimated by Cao et al. ([2025](#)) for four main crops globally from 1997-2015. I consider the ‘main’ crop in cells with multiple crops as the highest-yield crop in the cell. Annual net cell migration for 2000-2018 is estimated by Niva et al. ([2023](#)).

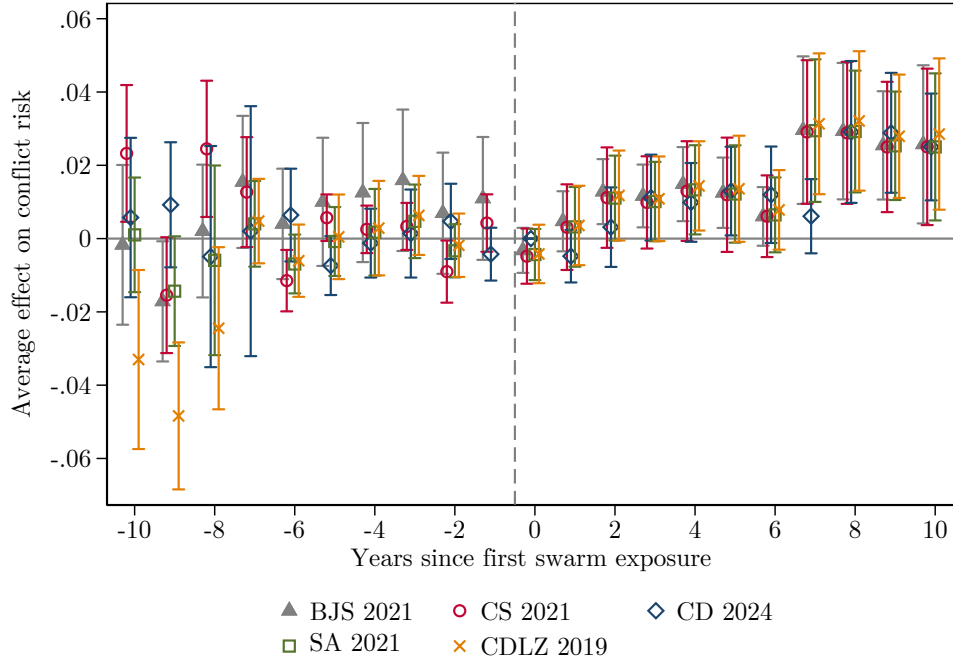
B Robustness

Table B1: Event study estimates of impact of swarm exposure on different conflict outcomes

	(1) Violent conflict (ACLED)	(2) Output conflict (ACLED)	(3) Factor conflict (UCDP)
pre10	-0.006 (0.011)	-0.003 (0.005)	-0.007 (0.010)
pre9	-0.024*** (0.008)	-0.009 (0.007)	-0.017** (0.007)
pre8	-0.006 (0.009)	-0.005 (0.007)	-0.006 (0.007)
pre7	0.004 (0.008)	-0.001 (0.006)	0.002 (0.009)
pre6	-0.007 (0.006)	-0.007 (0.006)	-0.002 (0.008)
pre5	0.000 (0.008)	-0.004 (0.006)	0.004 (0.010)
pre4	0.004 (0.009)	0.000 (0.006)	-0.002 (0.009)
pre3	0.005 (0.008)	0.000 (0.007)	0.003 (0.010)
pre2	-0.002 (0.007)	0.000 (0.006)	0.001 (0.008)
pre1	0.002 (0.007)	0.005 (0.007)	-0.001 (0.007)
tau0	0.000 (0.003)	0.002 (0.002)	0.001 (0.002)
tau1	0.003 (0.004)	-0.001 (0.003)	-0.001 (0.002)
tau2	0.009** (0.004)	0.007** (0.003)	0.006* (0.004)
tau3	0.009** (0.004)	0.007** (0.003)	0.004 (0.003)
tau4	0.015*** (0.004)	0.011*** (0.003)	0.003 (0.003)
tau5	0.013*** (0.004)	0.011*** (0.004)	0.008** (0.004)
tau6	0.005 (0.004)	-0.000 (0.003)	0.003 (0.003)
tau7	0.024*** (0.009)	0.018*** (0.005)	0.011* (0.007)
tau8	0.025*** (0.009)	0.021*** (0.006)	0.012*** (0.004)
tau9	0.021*** (0.007)	0.021*** (0.006)	0.004 (0.004)
tau10	0.027*** (0.009)	0.034*** (0.008)	0.007* (0.004)
Total annual precipitation (SDs)	0.004** (0.002)	0.002* (0.001)	-0.000 (0.001)
Max annual temperature (SDs)	0.012** (0.005)	0.005 (0.004)	0.005 (0.004)
Population (10,000s)	0.010*** (0.002)	0.009*** (0.002)	0.003*** (0.001)
Observations	295348	295348	295348

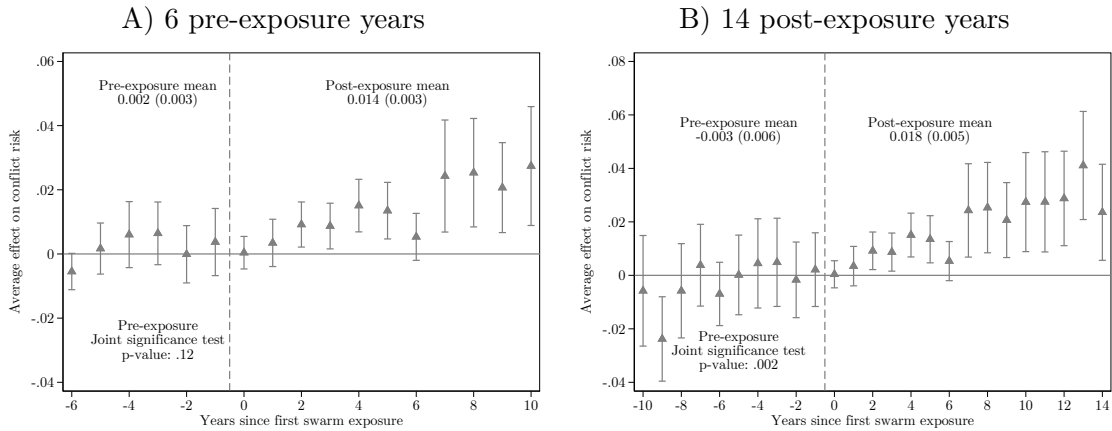
Note: The table shows the results of event study estimates of the impact of locust swarm exposure, with each column considering a different conflict outcome. Column 1 presents the results from [Figure 3](#), while columns 2 and 3 present the results from [Figure A1](#). Estimated impacts in each time period are weighted averages across effects for swarm exposure in particular years, calculated using the BJS estimator. Time period 0 is the year of first swarm exposure. Brackets represent 95% confidence intervals using SEs clustered at the sub-national region level. All regressions include country-by-year and cell fixed effects and controls for annual precipitation, maximum temperature, and population, and distance to capital-by-year fixed effects. Observations are grid cells approximately 28×28km by year.

Figure B1: Alternative locust swarm exposure event study estimators



The figure shows event study impacts of locust swarm exposure on the risk of any violent conflict estimated using different methods. ‘BJS 2021’ refers to the Borusyak et al. (2024) method first introduced in 2021, ‘CS 2021’ refers to Callaway and Sant’Anna (2021), ‘CD 2024’ refers to De Chaisemartin and d’Haultfoeuille (2024), ‘SA 2021’ refers to Sun and Abraham (2021), and ‘CDLZ 2019’ refers to Cengiz et al. (2019). Time period 0 is the year of first swarm exposure. Brackets represent 95% confidence intervals using SEs clustered at the sub-national region level. Observations are grid cells approximately 28×28 km by year. All regression include cell and year fixed effects and no controls, in contrast to the main specification which uses BJS with cell and country-by-year fixed effects and controls, due to constraints in including these additional controls in the Stata packages implementing some of the alternative estimators. I test the sensitivity of the main BJS estimates to different fixed effects and controls in Figure B5.

Figure B2: Impacts of exposure to locust swarms on violent conflict risk over different time horizons



Each panel replicates Figure 3 but changes the number of time periods included as indicated in the panel title. See the figure note for Figure 3 for more detail.

Table B2: Average impacts of exposure to locust swarms on violent conflict risk by estimator

Outcome: Any violent conflict event	(1)	(2)	(3)	(4)	(5)	(6)
	BJS 2021	CS 2021	CD 2024	SA 2021	CDLZ 2019	TWFE
Average post-treatment effect	0.014*** (0.005)	0.014*** (0.005)	0.013*** (0.004)	0.015*** (0.004)	0.016*** (0.005)	0.017*** (0.006)
Observations	295348	301664	301664	301664	301664	301664

Note: The table shows the results from estimating average long-term impacts of swarm exposure on violent conflict using different estimators. ‘BJS 2021’ refers to the Borusyak et al. (2024) method first introduced in 2021, ‘CS 2021’ refers to Callaway and Sant’Anna (2021), ‘CD 2024’ refers to De Chaisemartin and d’Haultfoeuille (2024), ‘SA 2021’ refers to Sun and Abraham (2021), ‘CDLZ 2019’ refers to Cengiz et al. (2019), and ‘TWFE’ refers to the two-way fixed effects specification shown in ???. As in Figure B1, all regressions include cell and year fixed effects and no controls, in contrast to the main specification which uses BJS with cell and country-by-year fixed effects and controls, due to constraints in including these additional controls in the Stata packages implementing some of the alternative estimators. The results in column 1 using BJS therefore differ from those in Table 1 column 1 which follow the main specification. The purpose of the table is to test sensitivity of the estimates to the choice of estimator. Observations are grid cells approximately 28×28km by year. SEs are clustered at the sub-national region level.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

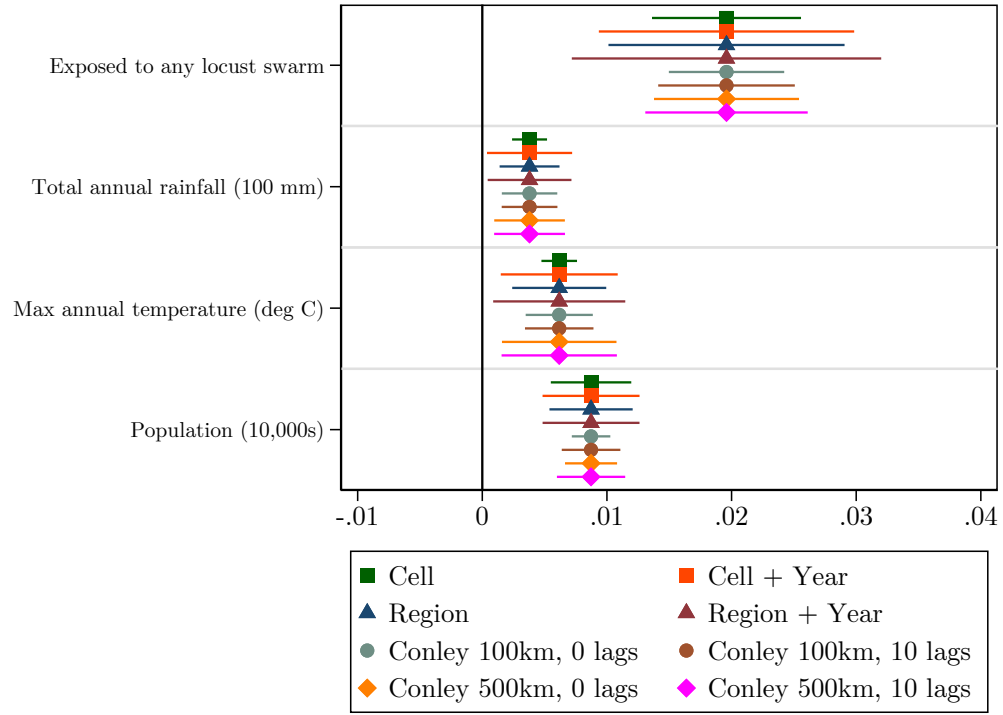
Table B3: TWFE average impacts of exposure to locust swarms on violent conflict risk by land cover

Outcome: Any violent conflict event	(1)	(2)	(3)
	All land	Land = Any crop or pasture land	Land = Any crop land
Exposed to any locust swarm	0.020*** (0.005)	0.005 (0.006)	0.006 (0.006)
Total annual precipitation (SDs)	0.004** (0.002)	0.002* (0.001)	0.002* (0.001)
Max annual temperature (SDs)	0.014** (0.005)	0.012** (0.006)	0.013** (0.006)
Population (10,000s)	0.009*** (0.002)	0.026** (0.011)	0.007 (0.006)
Exposed to any locust swarm × Land=1		0.017** (0.008)	0.021** (0.009)
Total annual precipitation (SDs) × Land=1		0.002 (0.002)	0.003* (0.002)
Max annual temperature (SDs) × Land=1		0.001 (0.004)	0.000 (0.005)
Population (10,000s) × Land=1		-0.018 (0.011)	0.002 (0.006)
Observations	301664	301664	301664
Outcome mean post-2004, no exposure	0.028	0.028	0.028
Country-Year FE	Yes	Yes	Yes
Cell FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes

Note: The table replicates the results in Table 1 but using a two-way fixed effects (TWFE) estimator. Column 1 includes no land cover interactions and the other two interact all right-hand side variables (except cell fixed effects) with cell land cover dummies. The ‘Land=1’ rows show the coefficients for the interaction of right-hand side variables with cell land cover dummies indicated in the column heading. The outcome mean for control cells is shown for post-2004 for comparison with exposure impacts in the period after the majority of swarm exposure occurred. All regressions include country-by-year, cell, and distance to capital by year fixed effects in addition to the controls shown. Observations are grid cells approximately 28×28km by year. SEs are clustered at the sub-national region level.

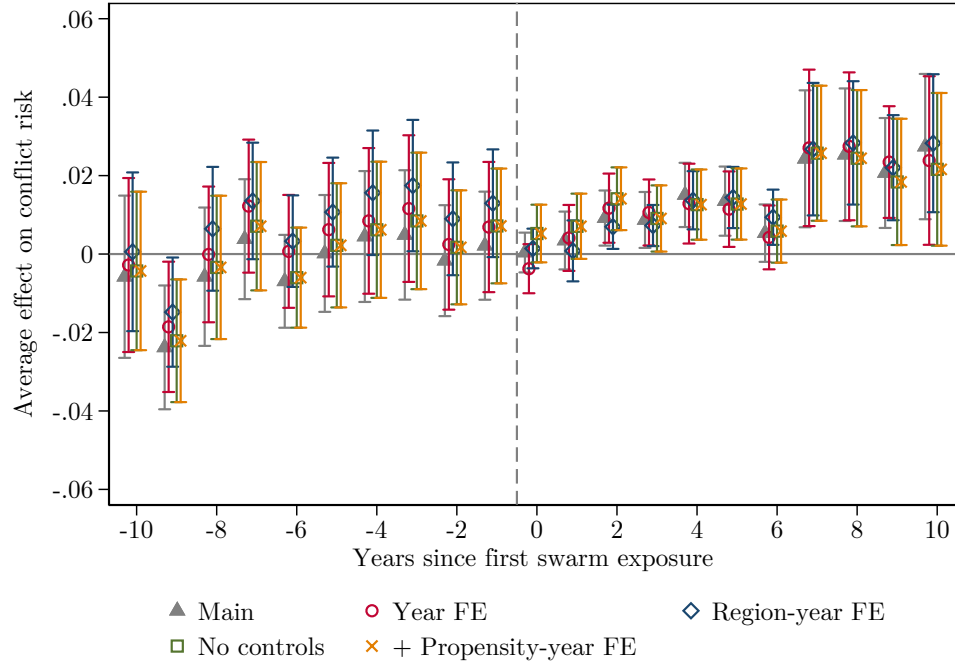
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Figure B3: Estimated coefficients from ?? with different SEs



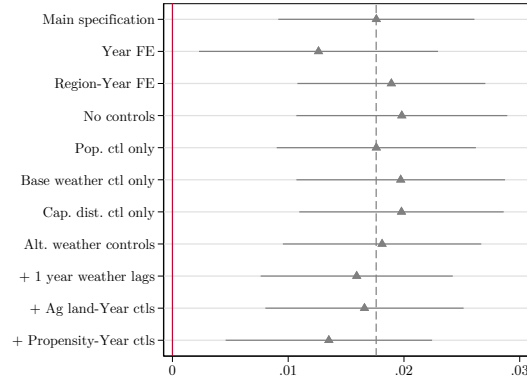
Note: The outcome variable is a dummy for any violent conflict observed. The figure shows 95% confidence intervals for estimates from [Table B3](#) column (1) applying different clustering for the SEs. Regions are clusters of sub-national administrative units constructed so that each one includes at least 32 grid cells. Observations are grid cells approximately 28×28 km by year. Regressions also include country-by-year and cell FE.

Figure B4: Sensitivity of locust swarm exposure event study results to different specifications



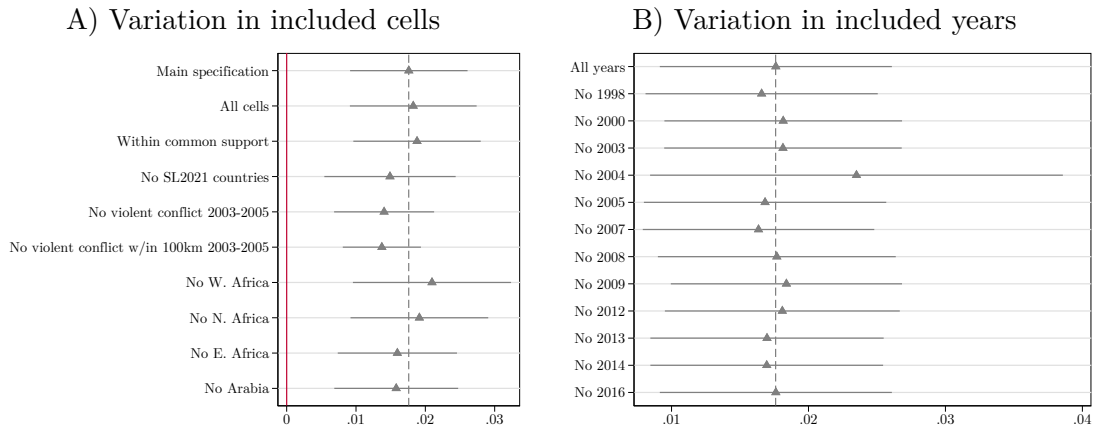
Note: Each event study replicates Figure 3 but changes some aspect of the specification as indicated in the legend. Time period 0 is the year of first swarm exposure. Brackets represent 95% confidence intervals using SEs clustered at the sub-national region level. Observations are grid cells approximately 28×28km by year. All regressions estimate dynamic effects of swarm exposure on violent conflict risk using the BJS estimator, and include country-by-year and cell fixed effects and controls for annual precipitation, maximum temperature, and population, and distance to capital by year fixed effects unless otherwise stated. The year and region-year FE specifications replace country-by-year FE with FE at these levels, where region indicates the same sub-national regions used in clustering the SEs. The propensity-year FE specification adds estimated swarm exposure propensity by year fixed effects; the estimation of these propensities is explained in the Empirical approach section.

Figure B5: Sensitivity of average impacts of locust swarm exposure on violent conflict risk to alternative specifications



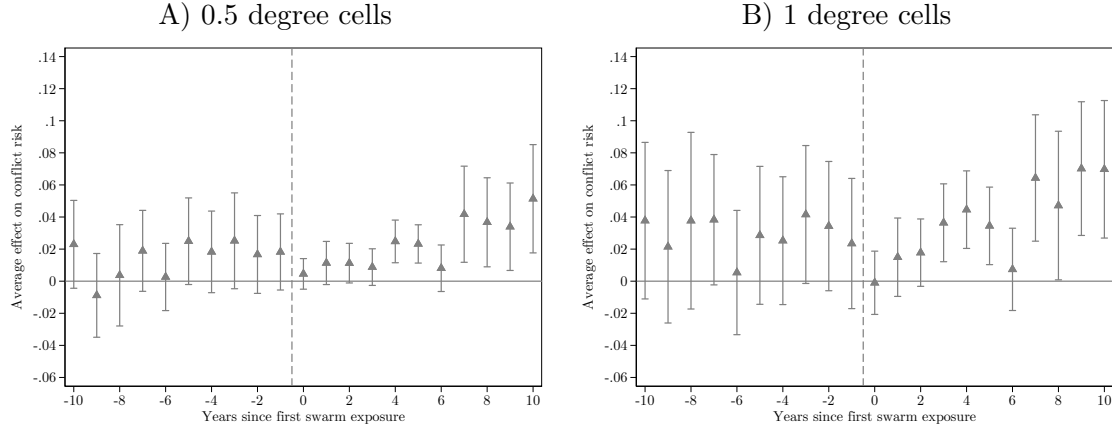
Note: Each estimate and 95% confidence interval is from a separate regression replicating Table 1 column 1 with a given modification. The dashed gray line indicates the main estimate from Table 1 column 1. The main specification includes controls for precipitation, temperature, and population at the cell-year level and cell, country-by-year, and distance to capital-by-year FE. Each coefficient is associated with a specific change in this specification. ‘Alt.’ weather controls replace the rainfall and temperature measures from WorldClim with measures from CHIRPS and ERA5, respectively. The bottom three specifications add additional controls: 1 year lags of precipitation and temperature, any agricultural land-by-year FEs, and estimated swarm exposure propensity-by-year FEs.

Figure B6: Sensitivity of average impacts of locust swarm exposure on violent conflict risk to alternative samples



Note: Each estimate and 95% confidence interval is from a separate regression replicating Table 1 column 1 with a given modification. The dashed gray line indicates the main estimate from Table 1 column 1. The main specification includes all years from 1997-2018 and excludes cells more than 100 km from any swarm report and outside the range of common support of estimated swarm exposure probability. Panel A shows results varying the set of included cells. I first vary whether cells more than 100 km from any swarm report and outside the range of common support of estimated swarm exposure probability are excluded. I then maintain the main sample restrictions but add additional restrictions. I drop countries where Showler and Lecoq (2021) report insecurity prevented some locust control operations during the sample period and cells that experienced violent conflict during the 2003-2005 locust upsurge which might have prevented locust reporting. Finally I drop countries in particular geographic regions. Panel B shows results from dropping individual years when locust swarm exposure events occurred.

Figure B7: Dynamic impacts of swarm exposure on violent conflict risk at different scales



Each panel replicates [Figure 3](#) estimating dynamic impacts of locust swarm exposure of the risk of any violent conflict event using the BJS method at different spatial scales. The main analysis uses 0.25° cells. When collapsing to larger cells I take the maximum of the swarm exposure and violent conflict variables and the mean of control variables across smaller cells within the aggregate cell. All regressions include country-by-year and cell fixed effects and controls for annual precipitation, maximum temperature, and population, and distance to capital by year fixed effects. SEs are clustered at the sub-national region level, which is assigned based on the region of the majority of 0.25° component cells.

Table B4: Average impacts of swarm exposure on violent conflict risk at different scales

Outcome: Any violent conflict event	(1)	(2)	(3)	(4)	(5)	(6)
Average effect of swarm exposure	0.018*** (0.004)	0.025*** (0.007)	0.028*** (0.011)	0.131*** (0.032)	0.118*** (0.033)	0.096*** (0.037)
Observations	301664	93940	30763	301664	93940	30763
Outcome mean post-2004, no exposure	0.028	0.067	0.128	0.028	0.067	0.128
Proportional effect of exposure	0.623	0.368	0.216			
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Cell FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Cell size (degrees)	0.25	0.5	1	0.25	0.5	1
Violent conflict outcome	Any	Any	Any	SDs	SDs	SDs

Note: The table replicates the results in [Table 1](#) column 1 at different spatial scales, as presented in the note to [Figure B7](#). Columns 1-3 show absolute effects on a dummy for any violent conflict while columns 4-6 show relative effects on the standard deviation in the probability of any violent conflict. The latter approach maintains comparability in effect sizes as the baseline risk of any conflict in a cell-year increase with cell size.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

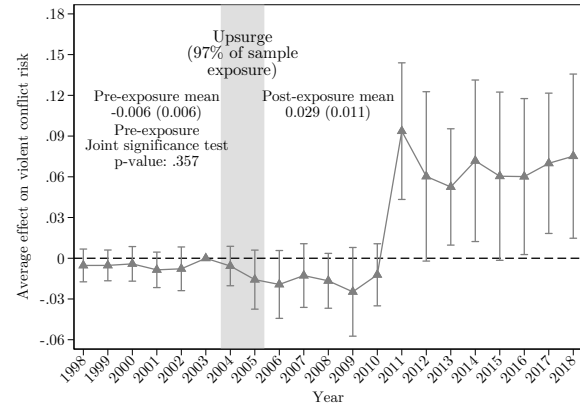
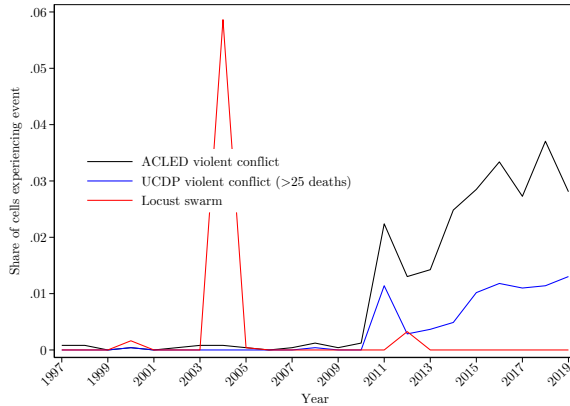
Table B5: Average impacts of direct and spillover exposure to locust swarms on violent conflict risk

Outcome: Any violent conflict event	(1) All swarms	(2) 2003-2005 upsurge swarms
Exposed to swarm	0.020*** (0.005)	
Exposed to swarm w/in 100km outside cell	0.001 (0.003)	
Exposed to swarm		0.018*** (0.005)
Exposed to swarm w/in 100km outside cell		0.011* (0.006)
Observations	301664	301664
Country-Year FE	Yes	Yes
Cell FE	Yes	Yes
Controls	Yes	Yes

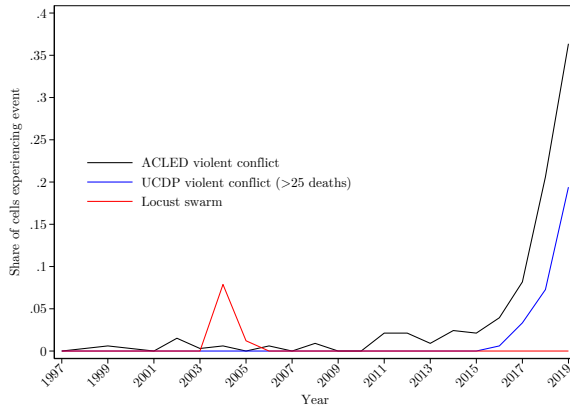
Note: The table presents results from estimating ?? but including a spillover swarm exposure variable based on the first year a swarm is outside the cell but within 100 km of a cell centroid. As with within-cell swarm exposure, spillover swarm exposure is coded as one in all years after the first nearby exposure in the study period. Column 1 considers all swarm exposure events while column 2 focuses on the 2003-2005 upsurge. All regressions include country-by-year and cell fixed effects and controls for annual precipitation, maximum temperature, and population, and distance to capital by year fixed effects. SEs are clustered at the sub-national region level. Observations are grid cells approximately 28×28km by year.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

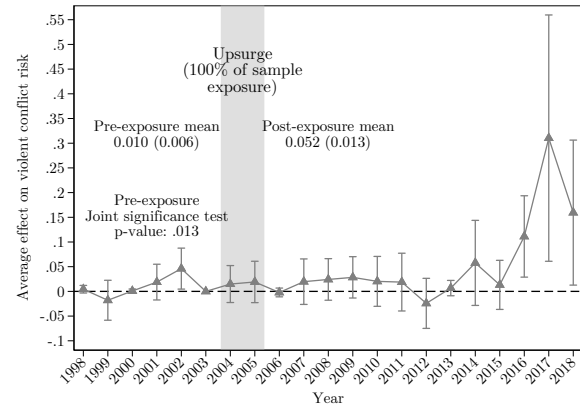
Figure B8: Changes in conflict environment and locust exposure event studies by country
A) Libya swarm and conflict trends
B) Libya upsurge event study



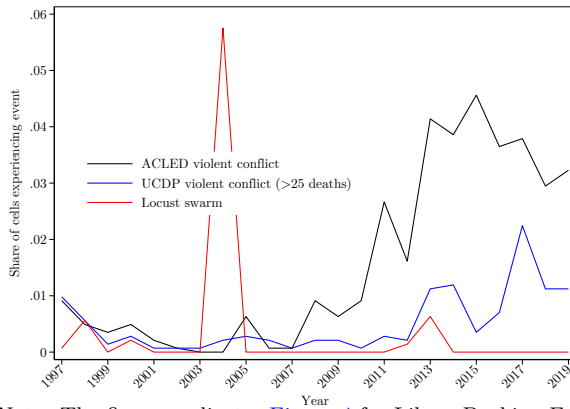
C) Burkina Faso swarm and conflict trends



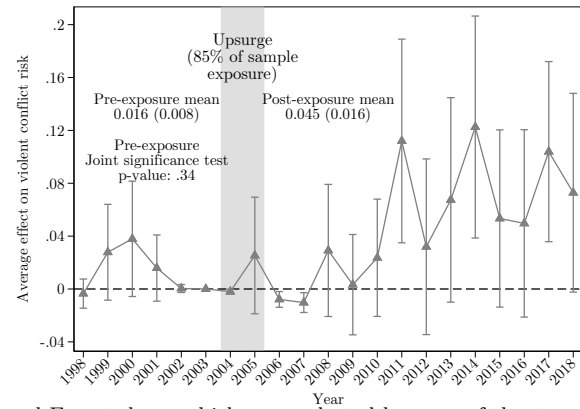
D) Burkina Faso upsurge event study



E) Egypt swarm and conflict trends



F) Egypt upsurge event study



Note: The figure replicates Figure 4 for Libya, Burkina Faso, and Egypt alone, which were selected because of the strong concentration of locust exposure during the 2003-2005 upsurge. Conflict in Libya began to increase in 2011 with the start of the Arab Spring and the subsequent civil war. Armed Islamist violence escalated in Burkina Faso starting in 2016, with many attacks by Al Qaeda and IS affiliates that established control over many rural parts of the country. Egypt underwent a revolution in 2011 as part of the Arab Spring and experienced a coup d'etat in 2013 and continued higher levels of insecurity.

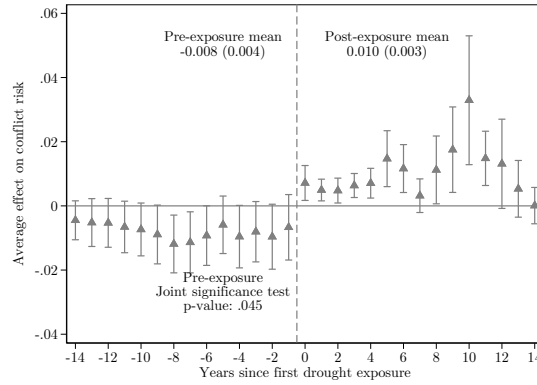
Table B6: Robustness of drought estimates to changing number of consecutive drought months

Outcome: Any violent conflict event	(1)	(2)	(3)	(4)	(5)	(6)
	5 month drought	5 month, Z = Any crop or pasture land	5 month, Z = Any conflict in country outside sub-region	6 month drought	6 month, Z = Any crop or pasture land	6 month, Z = Any conflict in country outside sub-region
Average effect of shock exposure	0.004** (0.002)			0.001 (0.002)		
Effect of exposure, Z=0		-0.002 (0.002)	0.000 (0.001)		-0.003* (0.002)	0.001 (0.001)
Effect of exposure, Z=1		0.008*** (0.003)	0.005** (0.003)		0.004 (0.003)	0.001 (0.002)
Observations	454986	454971	454986	455310	455300	455310
Outcome mean, no exposure	0.012	0.012	0.012	0.012	0.012	0.012
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Cell FE	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes

Note: The table reproduces columns 4-6 of [Table 5](#) changing the definition of what constitutes a drought. The main definition uses at least 4 consecutive months of drought (observed in 3.6% of cell-years), and the alternative definitions presented here use thresholds of 5 months (1.8% of cell-years) and 6 months (1.1%).

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

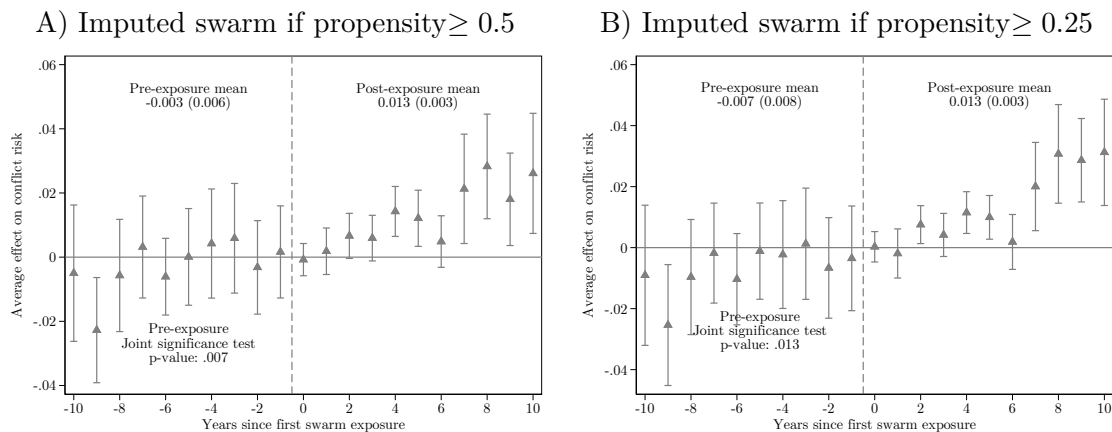
Figure B9: Event study impacts of drought exposure with 14 years of pre- and post-exposure estimates



The figures replicates [Figure 3](#) but changes the number of time periods included as indicated in the panel title. See the figure note for [Figure 3](#) for more detail.

C Simulations

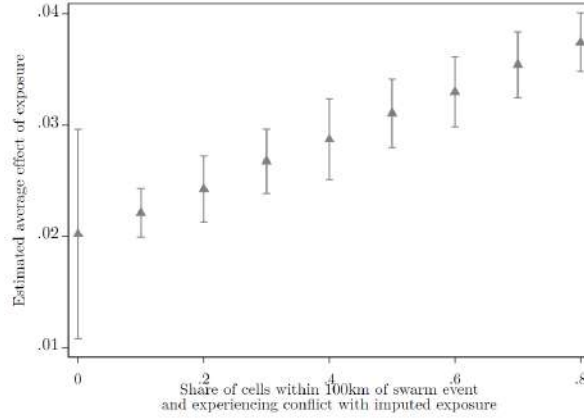
Figure C1: Impacts of exposure to locust swarms on violent conflict risk over time, imputing swarms in high-propensity areas



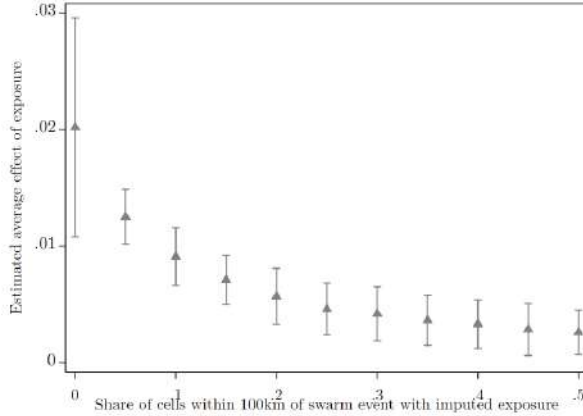
Each panel replicates [Figure 3](#) but changes the imputation of potential ‘missing’ locust swarms. In each case, I assign a cell as experiencing a locust swarm if its estimated propensity of exposure (over the whole study period) is above a certain level in any year where a locust swarm was reported within 100 km of the cell. I then define exposure based on the first year such a swarm is imputed, and estimate the event study as previously. See the figure note for [Figure 3](#) for more detail on the estimation.

Figure C2: Simulations of missing swarm observations

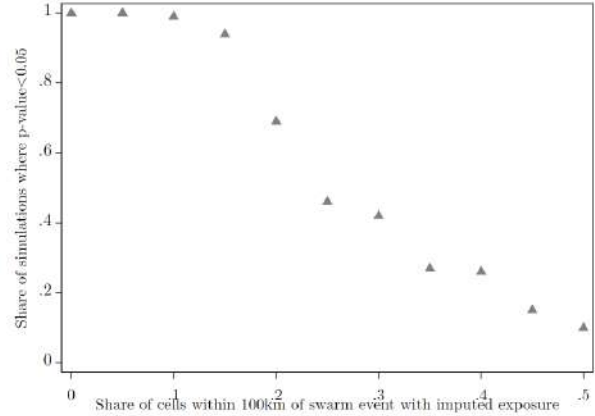
A) Betas, simulated swarms
in areas experiencing conflict
near swarm report



B) Betas, simulated swarms
in any area near swarm report



C) Rejection of H_0 , simulated swarms
in any area near swarm report



Note: The figures show the results from estimating ?? in simulations imputing the presence of increasing shares of unreported locust swarms in cell-years with a locust swarm reported within 100 km. For each share, I run 100 simulations randomizing which cell-years are imputed with an unreported swarm, recalculating the swarm exposure treatment variable, and estimating the average long-term impact of swarm exposure on violent conflict risk. In Panel A, I only impute swarms in cell-years both experiencing violent conflict and within 100 km of a reported locust swarm, to simulate effects of missing swarm reports in insecure areas. In Panels B and C, I impute swarms across all cell-years within 100 km of a reported locust swarm. Panels A and B report the average estimated effect across all simulations by share of imputed swarms, along with a 95% confidence interval for these estimates. Panel C reports the share of simulations where the p -value for the coefficient on swarm exposure is less than 0.05, by share of imputed swarms.

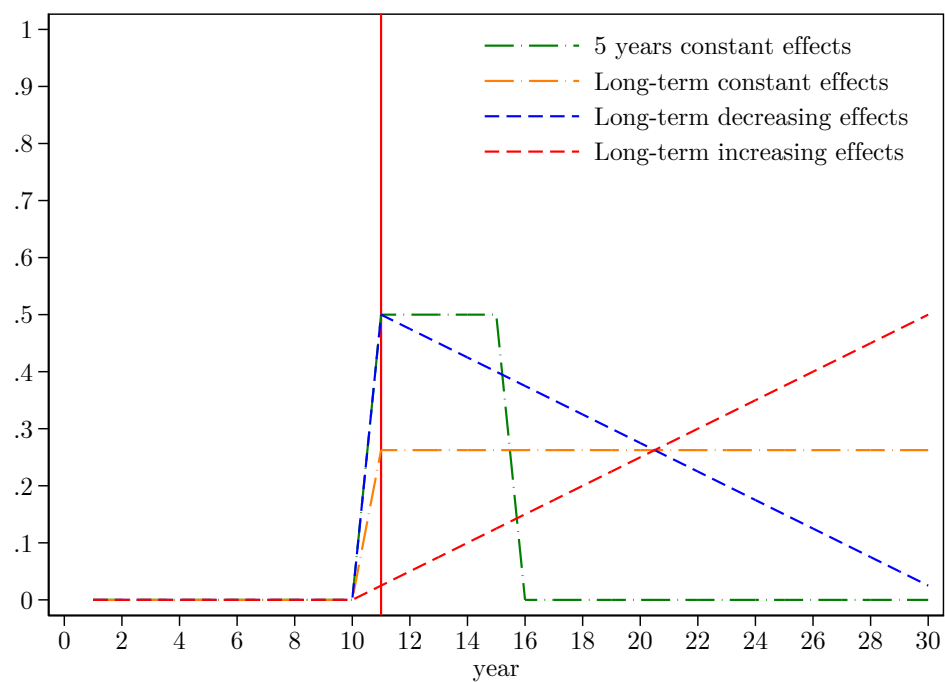
Table C1: Simulation: bias in estimates of short-term effects of shocks under different dynamic treatment effects

	(1) Immediate effect only	(2) Constant 5 year effects	(3) Constant long-term effects	(4) Decreasing long-term effects	(5) Increasing long-term effects
<i>A. True effects:</i>					
Treatment during current year	0.500*** (0.000)	0.500*** (0.000)	0.262*** (0.000)	0.500*** (0.000)	0.025*** (0.000)
Treatment 1 year lag	0.000 (0.000)	0.500*** (0.000)	0.262*** (0.000)	0.475*** (0.000)	0.050*** (0.000)
Treatment 2 year lag	0.000 (0.000)	0.500*** (0.000)	0.262*** (0.000)	0.450*** (0.000)	0.075*** (0.000)
Treatment 3 year lag	0.000 (0.000)	0.500*** (0.000)	0.262*** (0.000)	0.425*** (0.000)	0.100*** (0.000)
Treatment 4 year lag	0.000 (0.000)	0.500*** (0.000)	0.262*** (0.000)	0.400*** (0.000)	0.125*** (0.000)
<i>B. Estimated transitory effects:</i>					
Treatment during current year	0.500*** (0.000)	0.431*** (0.000)	0.091*** (0.000)	0.336*** (0.000)	-0.155*** (0.000)
Difference from actual treatment effect	0.000	0.069	0.171	0.164	0.180
<i>C. Transitory effects w/ 1st differences:</i>					
Treatment during current year	0.500*** (0.000)	0.250*** (0.000)	0.131*** (0.000)	0.263*** (0.000)	0 (.)
Difference from actual treatment effect	0.000	0.250	0.131	0.237	0.025
<i>D. Estimated lagged effects:</i>					
Treatment during current year	0.500*** (0.000)	0.500*** (0.000)	0.105*** (0.000)	0.380*** (0.000)	-0.170*** (0.000)
Treatment 1 year lag	0.000 (0.000)	0.500*** (0.000)	0.105*** (0.000)	0.355*** (0.000)	-0.145*** (0.000)
Treatment 2 year lag	0.000 (0.000)	0.500*** (0.000)	0.105*** (0.000)	0.330*** (0.000)	-0.120*** (0.000)
Treatment 3 year lag	0.000 (0.000)	0.500*** (0.000)	0.105*** (0.000)	0.305*** (0.000)	-0.095*** (0.000)
Treatment 4 year lag	0.000 (0.000)	0.500*** (0.000)	0.105*** (0.000)	0.280*** (0.000)	-0.070*** (0.000)
Differences from actual treatment effect	0.000	0.000	0.157	0.120	0.195
Observations	300000	300000	300000	300000	300000

Note: I simulate 10,000 observations across 30 periods, and randomly assign 20% to be treated in period 11. I define the outcome as taking a value of 0 for all units before period 11, and then vary the value based on different possible dynamic treatment effects. In column (1) treatment increases the outcome by 0.5 in the initial treatment period only. In column (2) treatment increases the outcome by 0.5 in each of the first 5 treatment periods. In column (3) treatment increases the outcome by 0.26 in all subsequent periods. In column (4) treatment increases the outcome by 0.5 in the first treatment period, but the effect *decreases* linearly over all subsequent periods. In column (5) treatment increases the outcome by 0.025 in the first treatment period, but the effect *increases* linearly over all subsequent periods. The evolution of the outcome over time is shown in [Figure C3](#). Patterns are similar with reversed signs if I simulate negative treatment effects. I estimate three treatment effect models for each scenario which make different assumptions about dynamic treatment effects. Panel A shows the results of event study estimates of the true effects for the first 5 treatment periods. Panel B shows the results of TWFE estimates that assume a transitory treatment that only affects the outcome in the initial treatment period. Panel C is similar to Panel B but includes four lagged treatment indicators, assuming treatment effects only persist for five periods. All regressions include period and unit fixed effects. SEs clustered at the unit level are in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Figure C3: Simulation: evolution of outcome under different dynamic treatment effects

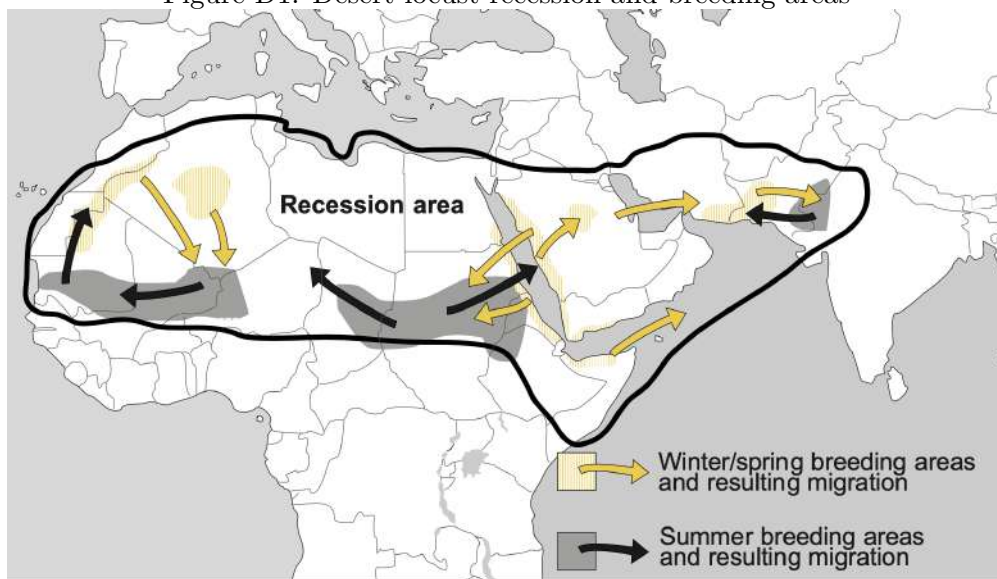


Note: This figure shows the evolution of the simulated outcomes as described in the notes to [Table C1](#).

D Desert locusts background

Desert locusts (*Schistocerca gregaria*) are a species of grasshopper always present in small numbers in desert ‘recession’ areas from Mauritania to India (Figure D1). They usually pose little threat to livelihoods but favorable climate conditions in breeding areas—periods of repeated rainfall and vegetation growth overlapping with the breeding cycle—can lead to exponential population growth. For example, the 2019-2021 locust upsurge persisted in large part because of repeated heavy precipitation out of season due to cyclones, prompting explosive reproduction (Cressman and Ferrand, 2021). The 2003-2005 upsurge was initiated by good rainfall over the summer of 2003 across four separate breeding areas. This was followed by two days of unusually heavy rains in October 2003 from Senegal to Morocco, after which environmental conditions were favorable for reproduction over the following 6 months (FAO and WMO 2016).

Figure D1: Desert locust recession and breeding areas



Source: Symmons and Cressman (2001). The recession area is the area in which desert locusts are most commonly found. The arrows indicate typical directions of downwind locust migration from breeding areas at different times of year.

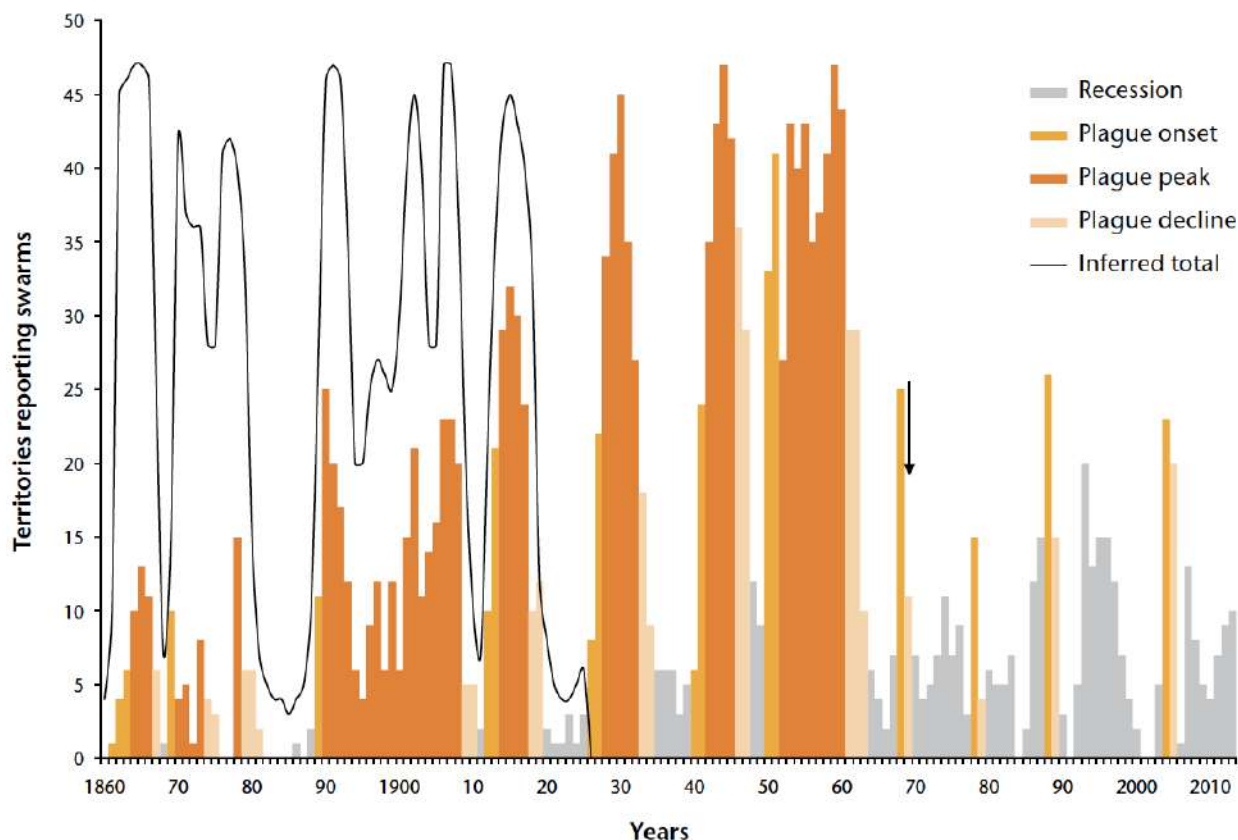
Unique among grasshopper species, after reaching a particular population density desert locusts undergo a process of ‘gregarization’ wherein they mature physically and form large bands or swarms which move as a cohesive unit (Symmons and Cressman, 2001). Locust bands may extend over several kilometers and alternate between roosting and marching, typically downwind (FAO and WMO 2016). In this paper, I focus my analysis on gregarious swarms of adult desert locusts and set aside observations of locusts in other phases.

Locust swarms form when bands of locusts remain highly concentrated when they reach the adult stage and become able to fly. Swarms vary in their density and extent (Symmons and Cressman, 2001). The average swarm includes around 50 locusts per m^2 with a range from 20-150, and can cover under 1 square kilometers to several hundred (Symmons and Cressman, 2001). About half of swarms exceed 50 km^2 in size (FAO and WMO 2016), meaning swarms typically include over a billion individuals.

The formation of swarms can lead to ‘outbreaks,’ where locusts spread out from their largely desert initial breeding areas. Locusts in swarms have increased appetites and accelerated reproductive cycles, and are thus particularly threatening to agriculture. The FAO distinguishes different levels of locust swarm activity (Symmons and Cressman, 2001). I use the terms ‘outbreak’ and ‘upsurge’ interchangeably to refer to any locust swarm activity. Few locust swarms are observed

outside of major outbreaks, as conditions favoring swarm formation tend to produce large swarms which reproduce and spread rapidly and are very difficult to control.

Figure D2: Desert locust observations by year



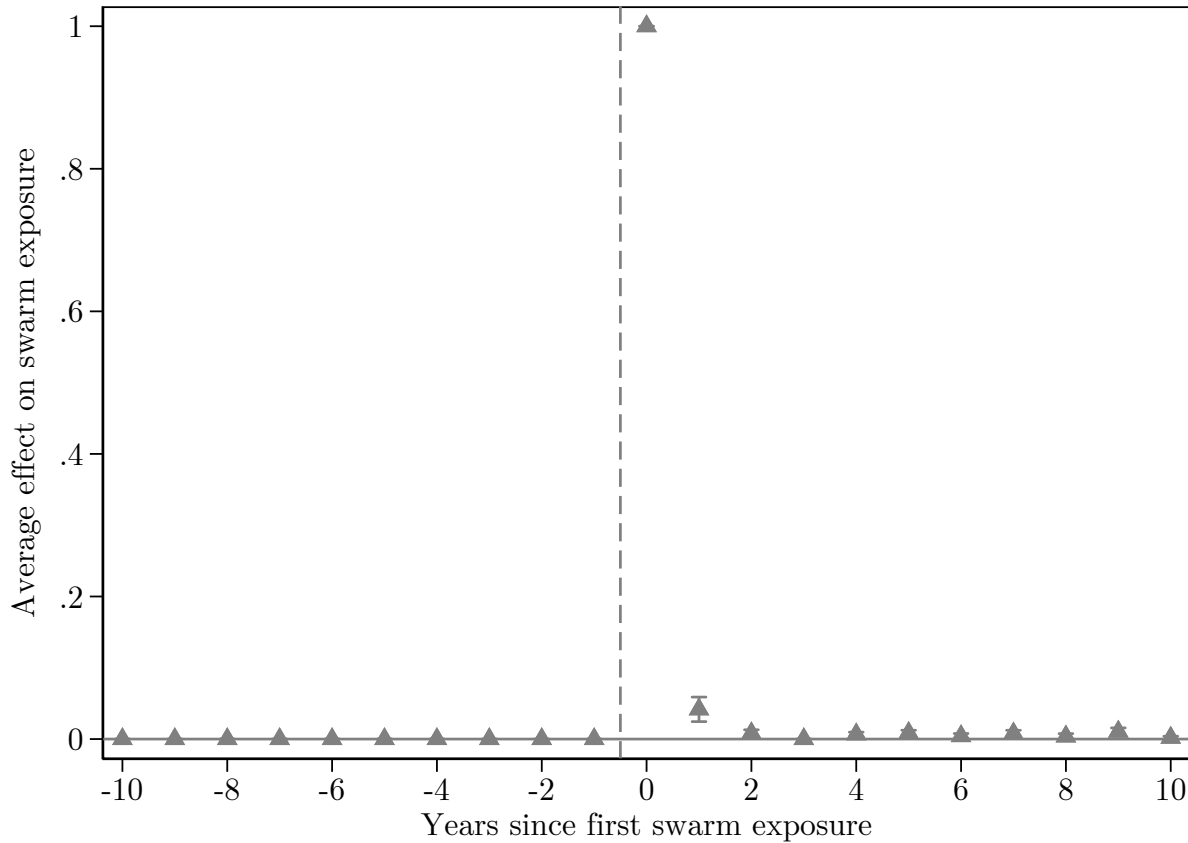
Source: Cressman and Stefanski (2016). Territories are generally individual countries.

The frequency of large-scale outbreaks has fallen since around the 1980s (Figure D2), in large part due to increases in coordinated preventive operations (Cressman and Stefanski, 2016). But climate change is expected to increase the risk of locust swarm formation and upsurges. Desert locusts can easily withstand elevated temperatures and the increased frequency of heavy rainfall events can create conditions conducive to population growth (Qiu, 2009; Youngblood et al., 2023).

As illustrated by Figure 1, locust swarms are not observed with any regularity over time or space. Desert locusts are migratory, moving on after consuming all available vegetation, and outside of outbreak periods are ultimately restricted to desert ‘recession’ areas. Unlike many other insect species, therefore, the arrival of a desert locust swarm does not signal a permanent change in local agricultural pest risk. Instead, the arrival of a swarm can be considered a locally and temporally concentrated natural disaster where all crops and pastureland are at risk (Hardeweg, 2001). Figure D3 illustrates how exposure to a locust swarm does not significantly affect the risk of exposure over the following years, consistent with exposure being a function of quasi-random variations in wind patterns and flight duration during swarm outbreaks. The only period in which cells exposed to locusts are more likely to experience a later locust swarm is in the period immediately after exposure, which reflects that many locust outbreaks last more than one year.

Locust swarms are migratory and fly downwind from a few hours after sunrise to an hour or so before sunset when they land and feed. Swarms do not always fly with prevailing winds and may wait for warmer winds. Small deviations in the positions of individual locusts in the swarm can also

Figure D3: Impact of swarm exposure on future exposure risk



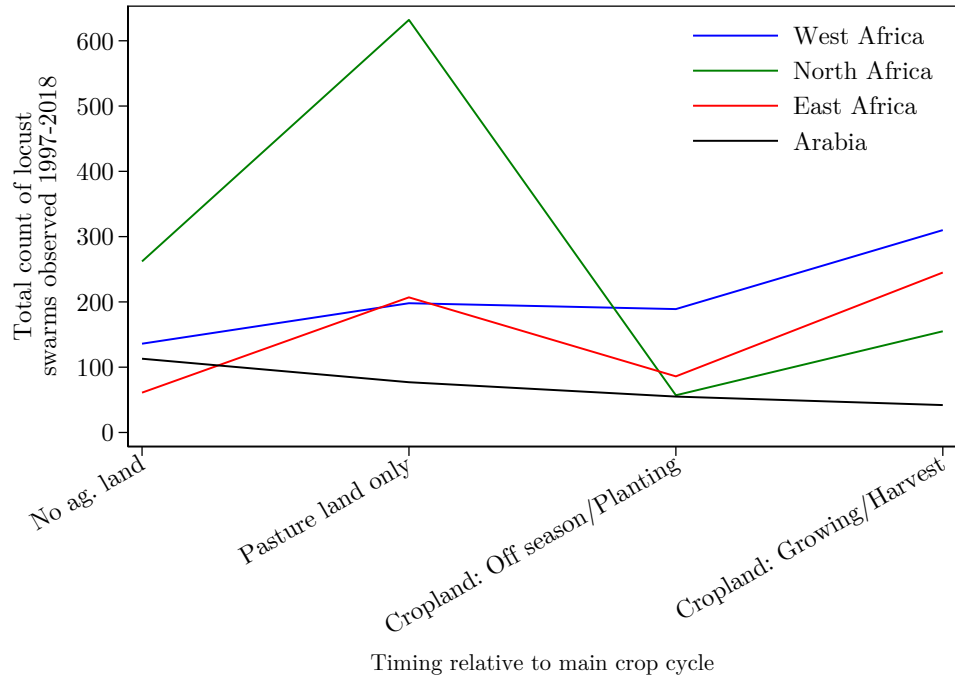
Note: The figure shows an event study of the impact of swarm exposure on the probability of having any locust swarm recorded in a subsequent year for the study period 1997-2018. By construction, none of the exposed areas had any locust swarm recorded in the years preceding their first exposure since 1990, and all had a locust swarm present in the year of exposure.

lead to changes in swarm flight trajectory, making their movements difficult to predict. Seasonal changes in these winds tend to bring locusts to seasonal breeding areas at times when rain and the presence of vegetation is most likely, allowing them to continue breeding (FAO and WMO 2016). Swarms are observed across different types of land cover and at different periods of the agricultural cycle, as shown in Figure D4.

Patterns in swarm movements lead to local variation in locust swarm exposure. After taking off, swarms fly for 9-10 hours rather than landing as soon as they encounter new vegetation. A swarm can easily move 100 km or more in a day even with minimal wind (Symmons and Cressman, 2001). Consequently, the flight path of a locust swarm will include both affected and unaffected areas, with the affected areas determined by largely by patterns of wind direction and speed over time from the initial swarm formation in breeding areas. Figure D5 illustrates the variation in areas affected by locust swarms over space around Mali. Swarm reports are densely clustered in the breeding areas in southern Mauritania where locust swarms reproduced in summer 2004. Outside of this area there is considerable variation in where swarms were reported, with distances between reported swarms over time consistent with typical flight distances.

These movement characteristics inform efforts to predict locust swarm movements, but these remain highly imprecise. The desert locust bulletins produced monthly by the FAO include forecasts of areas at risk of desert locust activity, but the areas described are quite large, often encompassing

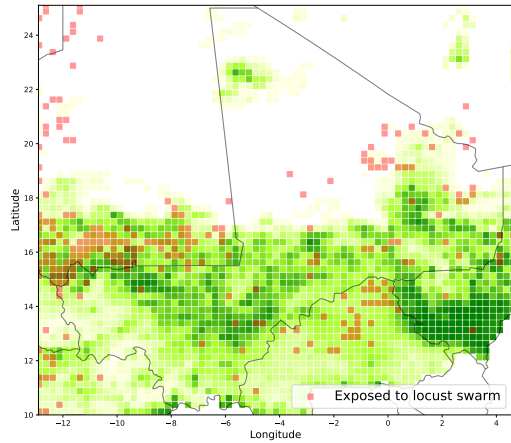
Figure D4: Variation in swarm timing and location during study period, by region



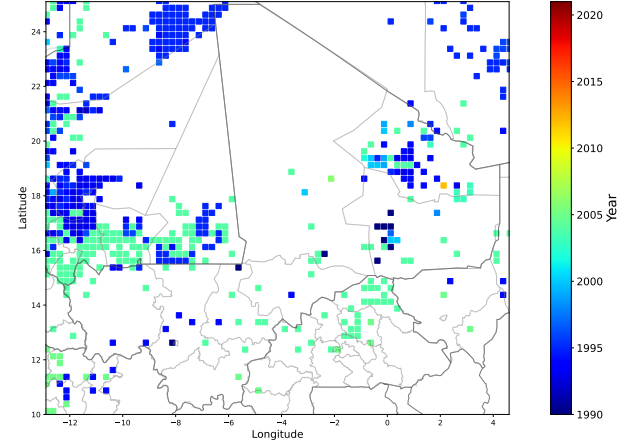
Note: The figure identifies the timing of swarm observations from 1997-2018 based on information on local crop calendars for major crops. This timing is not considered in cells with no cropland. These seasons are used to define timing of swarm exposure used in [Table 2](#).

Figure D5: Reports of locust swarms around Mali

A) Areas exposed to locust swarms since 1997



B) Year of first swarm exposure since 1990



Note: The figure illustrates the grid cells exposed to locusts swarms for the area around the country of Mali. Locust swarm reports are from the FAO Locust Watch database. Panel A overlays these reports on a map of the share of agricultural land area in each cell, while Panel B illustrates the timing of first exposure to locust swarms.

several countries in periods with increased swarms. While breeding regions and the broad areas at risk over different time periods can generally be predicted with some accuracy (Latchininsky, 2013; Samil et al., 2020; Zhang et al., 2019), predicting specific local variation in swarm presence remains a challenge due to the multiple factors influencing specific flight patterns (FAO and WMO 2016).

While desert locusts can exhibit some preferences over types of vegetation, avoiding woody

plants, dry vegetation, and some toxic plants in particular, swarms of locusts in their gregarious phase are less selective than solitary adults or nymphs (Despland, 2005; Despland and Simpson, 2005). The size of locust swarms and their polyphagous nature has led them to be considered the world’s most dangerous and destructive migratory pest (Cressman et al., 2016; Lazar et al., 2016). A small swarm covering one square kilometer consumes as much food in one day as 35,000 people and the median swarm consumes 8 million kg of vegetation per day (FAO, 2023), without preference for different types of crops (Lecoq, 2003).

The arrival of a locust swarm can therefore lead to the total destruction of local agricultural output (Showler, 2019). During the last locust upsurge in 2003-2005 in North and West Africa, 100, 90, and 85% losses on cereals, legumes, and pastures respectively were recorded, affecting more than 8 million people (Brader et al., 2006; Renier et al., 2015). Damages to crops alone were estimated at \$2.5 billion USD and \$450 million USD was required to bring an end to the upsurge (ASU, 2020). Over 25 million people in 23 countries were affected during the most recent 2019-2021 upsurge and damages were estimated to reach \$1.3 billion (Green, 2022), with control efforts—including treating over 2 million hectares with pesticides—estimated to have prevented over \$1 billion in damages (Newsom et al., 2021).

An important result of the local variation in locust swarm damages during outbreaks is that macro level impacts may be muted, since outbreaks occur in periods of positive rainfall shocks which tend to increase agricultural production in unaffected areas. Several studies find that impacts of locust outbreaks on national agricultural output and on prices are minimal, despite devastating losses in affected areas (Joffe, 2001; Krall and Herok, 1997; Showler, 2019; Thomson and Miers, 2002; Zhang et al., 2019). Chatterjee (2022) finds that wheat yields are 12% lower on average in Indian districts typically affected by desert locusts in years of locust outbreaks, in contrast to very large decreases in the specific areas exposed to locust swarms in those years.

Farmers have no proven effective recourse when faced with the arrival of a locust swarm, though activities such as setting fires, placing nets on crops, and making noise are commonly attempted. While these may slow damage they have little effect on locust population or total damages (Dobson, 2001; Hardeweg, 2001; Thomson and Miers, 2002). Desert locusts live 2-6 months and swarms continue breeding and migrating until dying out due to a combination of migration to unfavorable habitats, failure of seasonal rains in breeding areas, and control operations (Symmons and Cressman, 2001). The only current viable method of swarm control is direct air or ground spraying with pesticides (Cressman and Ferrand, 2021). These control operations do not prevent immediate agricultural destruction as they take some time to kill the targeted locusts, but will limit their spread. The 2003-2005 locust upsurge ended due to lack of rain and colder temperatures which slowed down the breeding cycle, combined with intensive ground and aerial spraying operations which treated over 130,000 km² at a cost of over US\$400 million (FAO and WMO 2016).

Desert locust control is most effective before locust populations surge, and the FAO manages an international network of early monitoring, warning, and prevention systems in support of this goal (Zhang et al., 2019). While improvements in desert locust management have been largely effective in reducing the frequency of outbreaks (as seen in Figure D2), many challenges remain. Desert locust breeding areas are widespread and often in remote or insecure areas. Small breeding groups are easy to miss by monitors, and swarms can migrate quickly. In addition, control operations are slow and costly, resources for monitoring and control are limited outside of upsurges, and the cross-country nature of the threat creates coordination issues. Insecurity may also limit locust control activities (Showler and Lecoq, 2021).

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